Trend2Line — Google Cloud Platform (GCP) Setup Guide

This guide walks you through setting up and deploying **Trend2Line** in the Google ecosystem from scratch, tailored specifically for **Windows** users who are new to GCP.

1. High-Level Architecture: How the Google Ecosystem Fits Together

Key Google Services Used:

1. **Google Cloud Project**: The container for all your cloud resources, billing, and permissions.
2. **Google AI Studio (Gemini)**: The AI engine powering the LLM stages (interpreting viral trends into product specs and writing audit narratives).
3. **Google Secret Manager**: A secure key-value store in GCP so you never hardcode API keys in code or Docker images.
4. **Google Cloud Run**: A fully managed serverless platform. It runs your Docker container, scales automatically (even down to 0 to save money), provides a secure HTTPS URL, and manages certificates automatically.
5. **Google Cloud Build**: Builds your Docker container directly inside GCP when you run `--source .`. **You do NOT need Docker installed on your Windows machine.**

2. Phase 1: One-Time Google Cloud & AI Setup (Web Browser)

Step 1: Create a Google Cloud Project

1. Open the Google Cloud Console.
2. Log in with your personal Google account.
3. Click the project dropdown at the top left and click **New Project**.
4. Name your project (e.g. `trend2line-prod`).
5. Note down your **Project ID** (e.g. `trend2line-prod-456789`). You will use this in all CLI commands.

Step 2: Link a Billing Account

> IMPORTANT
>
> Google Cloud requires an active billing account to use Cloud Run and Cloud Build. Google provides **$300 in free credits** for new accounts. Within the free tier limits, Cloud Run offers 2 million free requests per month, and Cloud Build offers 120 free build-minutes per day.

1. In Cloud Console, go to **Billing** from the left navigation menu.
2. Link your billing account or start the free trial.

Step 3: Get a Gemini API Key from Google AI Studio

1. Open Google AI Studio.
2. Click **Create API key**.
3. Select your Google Cloud project (or generate a default key).
4. Copy the API key string (starts with `AIza...`) and keep it safe.

3. Phase 2: Windows Tooling Setup

Run these commands in **Windows PowerShell** (run as Administrator if installing software):

Step 1: Install Google Cloud SDK (`gcloud` CLI)

Check if you have `gcloud` installed. If not, install via `winget`:

```
powershell
winget install Google.CloudSDK
```

(Alternatively, download the official Google Cloud SDK Windows Installer).

> WARNING
>
> After installing the Google Cloud SDK, **close and reopen your PowerShell terminal** so that `gcloud` is added to your environment `PATH`.

Verify installation:

```
powershell
gcloud --version
```

Step 2: Python Version Requirement (3.10+)

> NOTE
>
> The codebase uses modern Python type union syntax (`X | None`), which requires Python 3.10 or higher (Docker uses Python 3.12). If you wish to run the app locally, ensure Python 3.10+ is installed:

```
powershell
winget install Python.Python.3.12
```

4. Phase 3: Configure GCP from PowerShell

Open PowerShell and navigate to the project root directory:

```
powershell
cd "c:\Users\Swapnil Shinde\Downloads\trend2line\trend2line"
```

Step 1: Authenticate with Google Cloud

```
powershell
gcloud auth login
```

A browser window will open. Sign in and grant permissions.

Step 2: Set Your Active Project

Replace `YOUR_PROJECT_ID` with your actual Google Cloud Project ID:

```
powershell
$PROJECT_ID = "YOUR_PROJECT_ID"
gcloud config set project $PROJECT_ID
```

Step 3: Enable the Required Google Cloud APIs

Run this command to enable Cloud Run, Cloud Build, Artifact Registry, and Secret Manager:

```
powershell
gcloud services enable run.googleapis.com cloudbuild.googleapis.com
artifactregistry.googleapis.com secretmanager.googleapis.com
```

This may take 1-2 minutes to complete on a new project.

5. Phase 4: Configure Secrets & Permissions

Step 1: Store the Gemini API Key in Secret Manager

Replace `YOUR_AI_STUDIO_KEY` with your actual Gemini API key from Google AI Studio:

```
powershell
"YOUR_AI_STUDIO_KEY" | gcloud secrets create gemini-api-key --data-file=-
```

(If you ever need to update the key later, run: `"NEW_KEY" | gcloud secrets versions add gemini-api-key --data-file=-`)

Step 2: Grant Cloud Run Permission to Read the Secret

Cloud Run runs under the project's Compute Engine default service account. Grant it the `Secret Manager Secret Accessor` role:

```
powershell
$PROJECT_NUMBER = (gcloud projects describe $PROJECT_ID --format="value(projectNumber)")
gcloud secrets add-iam-policy-binding gemini-api-key `
  --member="serviceAccount:${PROJECT_NUMBER}-compute@developer.gserviceaccount.com" `
  --role="roles/secretmanager.secretAccessor"
```

6. Phase 5: Deploying Trend2Line to Cloud Run

We provide a dedicated PowerShell script for seamless Windows deployment.

Option A: Test Deploy in Mock Mode (No Gemini Calls, Instant & Safe)

Deploy with synthetic LLM responses to test your Cloud Run setup without consuming Gemini quota:

```
powershell
.\deploy.ps1 -ProjectId $PROJECT_ID -MockLlm 1
```

Option B: Deploy in Live Mode (Production with Gemini)

Deploy connected to Gemini Flash using your stored Secret Manager key:

```
powershell
.\deploy.ps1 -ProjectId $PROJECT_ID -GeminiModel "gemini-2.5-flash"
```

(You can also use another valid Flash model ID from AI Studio, e.g. `gemini-1.5-flash` *).*

What Happens During Deployment:

1. `deploy.ps1` runs to ensure hackathon compliance (Gemini-only).
2. It packages the source files (ignoring files listed in `.gcloudignore`).
3. Google Cloud Build creates the Docker container image inside GCP.
4. Cloud Run provisions a HTTPS service in region `asia-south1` (or your chosen region).
5. At the end, a public URL is output (e.g. `https://trend2line-xxxxxxxx-el.a.run.app`).

7. Phase 6: Testing & Verification

1. Check Health Endpoint

Open PowerShell or your browser and visit:

```
powershell
curl https://YOUR_SERVICE_URL/healthz
```

Expected response:

```
json
{"status":"ok","mock_llm":false}
```

2. Open the Web Dashboard

Open `https://YOUR_SERVICE_URL` in Chrome / Edge.

- Enter a trend name, e.g., **"Dubai chocolate"**.
- Click **Assess Trend**.
- Review the:
    - Wikipedia demand curve and half-life decay.
    - Bill of Materials & Allergen blast radius check.
    - Line compatibility verdict.
    - Verifier audit trail showing deterministic claim checks.

3. View Logs in Real Time

To monitor requests or debug errors directly from PowerShell:

```
powershell
gcloud run services logs tail trend2line --project $PROJECT_ID --region asia-south1
```

Or open **Google Cloud Console > Cloud Run > trend2line > Logs**.

8. Common Troubleshooting

Issue / Error	Cause	Fix
`'gcloud' is not recognized`	Cloud SDK not installed or terminal not restarted	Run `winget install Google.CloudSDK`, then close and reopen PowerShell.
`PermissionDenied: Secret Manager`	Service account missing secret accessor role	Re-run the IAM binding command in Step 5.2.
`Billing must be enabled`	GCP project lacks an active billing account	Link billing at console.cloud.google.com/billing.
`Container failed to start / crash`	Missing environment variable or secret name mismatch	Verify secret name is exactly `gemini-api-key` and check logs with `gcloud run services logs tail`.
Cold start latency (10-15s delay on first request)	Cloud Run scaling to 0 instances	Add `--min-instances 1` during demo presentations.